



HCF or LCM

Task A: HCF in a context

1) Aisha's little brother is sorting his 30 cars and 50 bricks into equal rows. What is the maximum number in each row if there are no rows with both cars and bricks?

$$30 = 2 \times 3 \times 5$$

$$50 = 2 \times 5^2$$

$$HCF = 2 \times 5 = 10$$

2) Jacob has 24 white, 42 red and 48 yellow flowers. What is the greatest number of identical bunches he can make that uses all of the flowers?

$$24 = 2^3 \times 3$$

$$42 = 2 \times 3 \times 7$$

$$48 = 2^4 \times 3$$

$$HCF = 2 \times 3 = 6$$

3) Lucas is making a model. He needs to cut his wire into the longest equal lengths possible. One piece is 420 cm and the other is 120cm. How long can he make each piece?

$$420 = 2^2 \times 3 \times 5 \times 7$$

$$120 = 2^3 \times 3 \times 5$$

$$HCF = 2^2 \times 3 \times 5 = 60$$

Task B: LCM in context

1) Jun goes swimming every 4 days and he goes to the cinema every 15 days. Which is the first day that he does both?

$$4 = 2^2, 15 = 3 \times 5, LCM = 2^2 \times 3 \times 5 = 60$$

The first day he does both is the 60th day

2) Jacob is baking he can bake 12 cupcakes and 20 biscuits at a time. He needs to make the same number of each. What is the smallest amount he can make?

$$12 = 2^2 \times 3, 20 = 2^2 \times 5, LCM = 2^2 \times 3 \times 5 = 60$$

The smallest amount he can make is 60 of each

3) Sam was making sandwiches for a party. Bread comes in loaves of 24 slices, cheese in packs of 15 slices and ham in packs of 40 slices. Each sandwich has 2 slices of bread and one slice of cheese and ham. What is the fewest of each she needs to buy to ensure no wastage?

Each sandwich uses 2 slices of bread so you can make 12 from 1 loaf.

$$12 = 2^2 \times 3, 15 = 3 \times 5, 40 = 2^3 \times 5, LCM = 2^3 \times 3 \times 5 = 120$$

Sam has to buy 120 of each item.



Task C: Whether to use HCF or LCM based on the context

1)

a) Two people race up some steps. One goes up 2 at a time and the other 3 at a time. Which is the first step they both step on?

LCM

b) Aisha is organising a charity quiz. She either has 84 or 90 people attending. What is the largest number of people in each team? (all teams must be the same size)

HCF

c) Grandad has baked a cake. His grandchildren are coming to visit, it could be 3, 5 or all 6. What is the smallest number of pieces he can cut the cake into so they each get the same amount of cake?

LCM

2)

a) Aisha is organising a charity quiz. She either has 84 or 90 people attending. What is the largest number of people in each team? (all teams must be the same size)

$$84 = 2^2 \times 3 \times 7$$

$$90 = 2 \times 3^2 \times 5$$

$$HCF = 2 \times 3 = 6$$

b) Grandad has baked a cake. His grandchildren are coming to visit, it could be 3, 5 or all 6. What is the smallest number of pieces he can cut the cake into so they each get the same amount of cake?

3

5

$$6 = 2 \times 3$$

$$LCM = 2 \times 3 \times 5 = 30$$

c) A red bus leaves the station every 3 minutes, a blue one every 8 minutes and a green one every 11 minutes. If they all leave at 09:00 at what time do they all leave at the same time again?

$$3 \quad LCM = 2^3 \times 3 \times 11 = 264 \text{ minutes}$$

$$8 = 2^3 \quad 264 \text{ minutes} = 4 \text{ hours } 24 \text{ minutes}$$

11 *They will leave at the same time again at 13:24*

